

Honeypot Router for routing protocols protection

Abdallah Ghourabi
gh.abdallah@gmail.com

Tarek Abbas
Tarek.abbes@isecs.rnu.tn

Adel Bouhoula
Adel.bouhoula@supcom.rnu.tn

Department of Computer Science and Networks, Higher School of Telecommunications SupCom,
2083 Cité El Ghazala, Tunisia

Abstract— Routing protocols are essential for interconnecting networks; however they may enclose several vulnerabilities that can be exploited by malicious attackers. For example, an attacker may send forged packets to a router with the intention of changing or corrupting the routing table, which in turn can reduce the network connectivity and degrade the router functionalities. To prevent and detect such attacks, several security techniques are available like firewall, authentication mechanisms and intrusion detection system (IDS). Nevertheless these security methods encounter some problems, especially when dealing with new attacks. Relying on additional security principles seems to be important to well protect network connectivity offered by routers.

In this paper, we propose using honeypot to protect routing protocols. Honeypot is particularly useful to attract attackers, driving them away from real routers and allowing the administrators to be aware about intrusion attempts on their networks and the employed techniques that can be recent. Our solution (Honeypot Router) is to deploy a honeypot playing the role of a router. The honeypot is based on routing software called Quagga and other tools for traffic capture and analysis. The entire solution supervises all routing traffic, so it detects and studies new attacks against routing protocols (RIP, OSPF and BGP).

Index Terms— Honeypot Router, routing protocol, routing attack detection, routing protocol protection.

I. INTRODUCTION

ROUTING protocols are used in the implementation of routing algorithms to ease the exchange of routing information among networks, allowing routers to build routing tables dynamically. These protocols are classified into two types depending on the size of the network that they serve: intra-domain (RIP, OSPF, etc.) and inter-domain (BGP).

Routing attacks represent serious threats in reality. An enterprise routing infrastructure can be easily attacked with man-in-the-middle and other attacks designed to corrupt or change the routing tables like traffic redirection, traffic sent to a routing “black hole”, router denial-of-service (DoS), routing protocol DoS and unauthorized route prefix origination. Securing routing protocols becomes very indispensable.

Routing security has received varying levels of attention over the past several years and has recently begun to attract more attention. Several researchers have proposed more secure protocols [14, 4, 9, 17]. Most of them strengthen the protocol mechanism by using standard digital signatures to authenticate routing update messages. The authentication is not very effective since it can be subject to some cryptanalysis attacks.

Another approach is to supervise the protocols execution in order to detect attacks scenarios and quickly undertake the appropriate countermeasures [7, 11, 1]. The technique presumes having a detailed view of the routing protocols in order to identify attacks by a bad state transition sequence or patterns. To understand attacks’ nature and hackers’ methods and tools, a complementary approach is needed. The idea is to use honeypots.

A honeypot is a computer system voluntarily vulnerable to one or more known threats, deployed on a network for the purpose of logging and studying attacks on the honeypot. These systems may be made purposely insecure in order to lure attackers to study their techniques, tools, and motivations. Despite the continuing evolution of this technique, honeypots present some limits, especially for performing/simulating routing services. We propose in this paper a solution which takes advantages of honeypots in order to detect and study the attacks against routing protocols. Our solution (Honeypot Router) is to deploy a honeypot playing the role of a router. The honeypot is based on the routing software Quagga and other tools for traffic capture and analysis. It supervises the routing traffic, so it can detect and study new attacks against routing protocols (RIP, OSPF and BGP).

The remaining parts of the paper are organized as follows: Section II reviews honeypots, their types and their relationship with routing. Section III outlines some serious vulnerabilities found in RIP, OSPF and BGP protocols. Section IV reviews related works regarding routing protocol security. In Section V we present our Honeypot Router solution. We describe in Section VI two experiments executed on our Honeypot Router to analyze attacks against RIP and OSPF protocols. Finally, we conclude the paper in Section VII

II. HONEYPOT USEFULNESS, CLASSES AND POSITION REGARDING ROUTING PROTOCOL

In recent years, honeypots are increasingly deployed in computer networks. They can collect useful information to resolve several security issues [15].

A. Presenting Honeypots

“A honeypot is an information system resource whose value lies in unauthorized or illicit use of that resource.” [15]. A honeypot is a computer system voluntarily vulnerable to one or more known threats, aimed at attracting attackers to explore their attack strategies. The various actions to be undertaken by the attackers will be recorded in a journal and will be analyzed

to understand strategies pursued by hackers to achieve their ends.

Honeypots store collected information in log files. This information is used to learn new techniques, tools and motivations of hackers to better protect the production systems against attacks. Such honeypots are called *research honeypots*. Another reason for the deployment of honeypots is the network protection. Here we speak about *protection honeypots*.

There are two honeypots classes according to the interaction they provide to attackers: *low-interaction* (for example Honeyd [19]) and *high-interaction* honeypots (HoneyNet [20]).

B. Honeypot and routing

Honeypots come in many shapes and sizes and are available to mimic lots of different kinds of applications and protocols [12]. Besides traditional honeypots, we point out client honeypots that search malicious servers; typically this has centered on web servers that deliver client-side browser exploits. Honeypots are also conceived for specific applications and protocols such as SMTP, PHP and HTTP [18] and for wireless environment [13]. However, despite this diversity, there is still a need to emulate routing services in order to supervise the routing infrastructure and protect the network connectivity.

Frequently, a honeypot is deployed as an end system, so it cannot capture routing traffic. In this way we cannot investigate attacks against routing protocols. Among the few honeypots that allow a router simulation, you find Honeyd. The software emulates a Cisco router but it lacks some rudiments features such as emulating routing protocols. In addition, Honeyd is a low-interaction honeypot, so it cannot interact heavily with the attacker, which prevents us from collecting much information about routing attacks.

The lack of a honeypot solution in the routing infrastructure leads us to design a Honeypot Router. This equipment constitutes a basis for studying attacks against routing protocols. We apply our solution on RIP, OSPF and BGP protocols which we investigate their vulnerabilities in the next section.

III. ROUTING PROTOCOLS VULNERABILITIES

A. RIP vulnerabilities

RIP (Routing Information Protocol) is an intra-domain distance-vector routing protocol [6]. It employs the hop count as a routing metric. RIP is vulnerable to several types of attacks like “blackhole” attack, false routes announcement and Denial of service (DoS) [3].

B. OSPF vulnerabilities

Open Shortest Path First (OSPF) routing protocol is an interior gateway protocol (IGP), which is designed to work within an autonomous system (AS). OSPF uses the link-state information to build a routing table by executing the shortest path first (SPF) algorithm. Each router running OSPF periodically diffuses a link-state advertisement (LSA) throughout the AS to announce its neighbours and the corresponding routing metrics. Although OSPF offers some

authentication mechanisms to exchange information between routers, it is possible to bypass it. There are three well-known attacks against OSPF protocol: Seq++ attack, MaxAge and MaxSeq attack [1].

C. BGP vulnerabilities

Border Gateway Protocol (BGP) is used to exchange routing information across Internet. BGP is considered as an inter-domain routing protocol. It maintains networks IP addresses or “prefixes” in a routing table to represent network reachability toward autonomous systems (AS). It is viewed as a path vector protocol. BGP is the only routing protocol using TCP as a transport protocol. Thus, it is more difficult to intercept messages and inject false data. Nevertheless the protocol remains vulnerable to some attacks: BGP Hijacking, BGP session reset, DoS attack, etc. [10].

IV. RELATED WORK

Several works have been done to secure routing protocols. In the context of RIP, Smith, Murthy and Garcia-Luna-Aceves [14] described a scheme to secure distance-vector routing protocols by using predecessor information. Hu, Perrig and Johnson [3] proposed several efficient mechanisms using one-way hash chains and authentication trees for securing distance vector routing protocols. Mittal and Vigna [7] proposed a sensor-based intrusion detection method. Their technique relies on the network topology to automatically generate signatures used by sensors to detect routing attacks.

To secure OSPF protocol, Murphy and Badger [8] proposed a preventive approach using cryptography, specifically, digital signatures to protect the integrity of LSAs (Link-State Advertisement). Chang et al. [1] provided a real time intrusion detection system called JiNao to supervise link-state routing protocol as OSPF. JiNao uses finite state machines to detect attacks targeting this protocol.

Moreover, several proposals have been presented to secure BGP protocol. Kent et al. [4] described S-BGP (Secure Border Gateway Protocol) which proposes the use of centralized PKIs for authenticating AS numbers and IP prefix ownership. A group of researchers from Cisco Systems proposed soBGP (Secure Origin BGP) [9] for validating the correctness and authorization of the data carried within BGP, and also for preventing the advertisement of unauthorized prefixes. Another improvement to BGP called psBGP (Pretty Secure BGP) was proposed by Wan et al. [17].

To reinforce the security of the routing protocols, we need an additional approach able to detect new attacks and understand the techniques and methods used by attackers. Our idea is to use a honeypot as a complement solution to existing security tools. In fact, honeypots have proven their effectiveness in detecting new attacks (0-day attack), thanks to an ongoing analysis of the network traffic [15]. However, as already mentioned there is a lack in the deployment of honeypots as routers and routing protocols. Therefore we propose to deploy a Honeypot Router that detect and investigate attacks against routing protocols. *The Honeypot Router is a new idea which can be useful to strengthen the*

security of routing protocols. To the best of our knowledge there are no researches that offer honeypots particularly concerned with routing protocols. We develop the solution in the next section.

V. HONEYPOT ROUTER FOR ROUTING PROTOCOLS PROTECTION

Our solution is to deploy a honeypot as a router that captures and monitors ingress and egress traffic to detect and study attacks targeting routing protocols.

A. Description

The router is based on Quagga [21], an open source routing software (successor of Zebra) which can transform a Unix machine into a real router by implementing major routing protocols such as RIP, OSPF and BGP. Our Honeypot Router is deployed on a Unix machine, which contains the Quagga software and other monitoring and data capture tools. These tools help us to analyze the data sent to the router in order to detect attacks and learn more about the hacking techniques. The tools list contains: Quagga [21], Wireshark [22], Argus [23], Snort Inline [24], Psad [25] and P0f [26].

The purpose of the honeypot deployment is to detect and analyze attacks that target the router. To ensure a smooth deployment of the Honeypot Router, it is preferable to avoid virtual machines since it can be easily detected by attackers [2]. It should also be noted that to attract attackers, it is better to deploy the Honeypot Router in interesting places (in ISPs networks for example).

B. Architecture

Our Honeypot Router can be decomposed into 4 functional blocks essential to provide the basic tasks and an extra router behavior for the Honeypot (Figure 1).

- **Routing simulation:** the purpose of this block is to convince attackers that they are interacting with a real router. Therefore, honeypot must execute routing services and support routing protocols. To transform our honeypot into such device, we install Quagga. This software converts a Unix machine into a real router by implementing the principal protocols such as RIP, OSPF and BGP.
- **Traffic capture:** this block captures network traffics in order to trace attackers' activities. For this purpose we use *libpcap* and *tcpdump*.
- **Traffic analysis:** captured data will be filtered then analyzed to ensure 4 activities: port scanning detection, OS/network fingerprinting detection, flow analysis and routing protocols traffic monitoring. We incorporate into our honeypot 4 tools: Wireshark, Argus, P0f and Psad. Wireshark analyzes captured packets, especially routing protocols packets. Argus generates information about the flows (source and destination addresses and ports, protocol, start and end of the flow...). Psad detects port scans. P0f is used as an advanced passive OS/network fingerprinting utility.

- **Traffic control:** this block prevents the exploitation of our honeypot to launch attacks and corrupt other systems. Traffic Control block mitigates this risk. It executes the Intrusion Prevention System (IPS) "Snort_inline" that relies on Snort IDS signatures to analyze packets traversing the Honeypot.

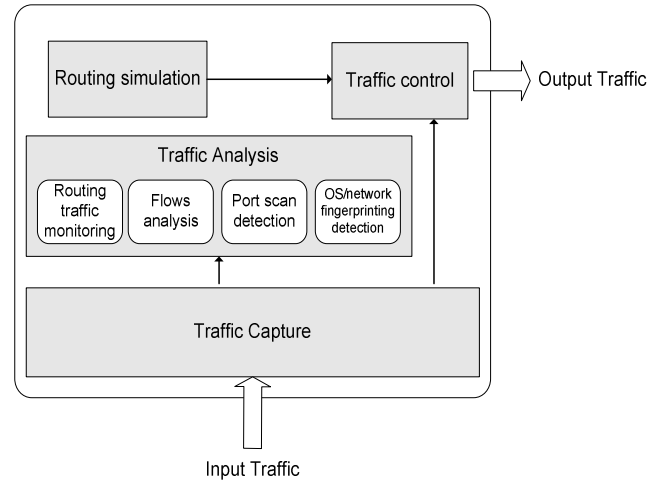


Figure 1. Honeypot Router Architecture

VI. EXPERIMENTAL ENVIRONMENT AND RESULTS

In order to test the functioning of our Honeypot Router, we execute two attacks that target RIP and OSPF protocols.

A. Experiment 1: Attack against RIP protocol

For the execution of our experiment, we placed our Honeypot Router in a test network. The routing table of the Honeypot Router is schematized in Figure 2.

The purpose of the attack is to inject a RIP packet version 2 in the Honeypot Router (100.0.0.1) by an attacker (100.0.0.3). The forged packet claims that destination 192.168.100.0/24 is reachable with a lower cost (1). This route change in the Honeypot Router can cause instability in the network.

Destination	Gateway	Genmask	Ind	Metric	If
192.168.100.0	101.0.0.1	255.255.255.0	UG	3	eth0
192.168.200.0	100.0.0.2	255.255.255.0	UG	2	eth1
102.0.0.0	101.0.0.1	255.0.0.0	UG	2	eth0
101.0.0.0	0.0.0.0	255.0.0.0	U	0	eth0
100.0.0.0	0.0.0.0	255.0.0.0	U	0	eth1

Figure 2. Routing table of the Honeypot Router before the attack.

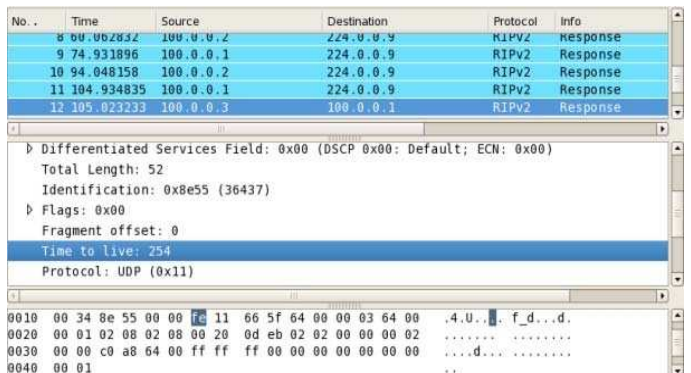


Figure 3. Malicious RIP packet captured by Wireshark.

Using Wireshark, we detected the suspect RIP packet (Figure 3). The latter is sent from the address 100.0.0.3 in unicast towards our router (100.0.0.1). By analyzing the packet, we notice that it is sent with a TTL (Time to live) equal to 254, which should not be the case for a neighbour router. In fact, a packet sent by a neighbour router must have a TTL equal to 1, which brings us to conclude that this packet is malicious.

B. Experiment 2: Attack against OSPF protocol

In the same network test we suppose that a malicious router tries to attack our Honeypot Router. To carry out the attack, the idea is to intercept a LSA and to modify its age to maximum (3600 s), then to re-inject it from the malicious router to the Honeypot Router. This LSA has the same sequence number but a maximum age, what leads the Honeypot Router to eliminate it from its data base. This attack is called MaxAge; it can cause a denial of services in the network.

After the attack, in the routing table of the Honeypot Router, we notice that the attack caused the deletion of two routes (192.168.100.0/24 and 102.0.0.0/8). By means of Wireshark we captured a suspect LSA packet (see Figure 4). This packet is sent by the address 101.0.0.1 “unicast” towards our router (101.0.0.2), what is not the case for OSPF who sends its packets “multicast” towards the address 224.0.0.5. Thus we conclude that the packet was sent in a particular way. In addition, the packet is sent with a TTL equal to 5 (the TTL of this kind of packet must be equal to 1). By examining the packet contents (Figure 4), we notice that the age associated with the sequence of this LSA is equal to 3600 (the maximal age); the purpose of its issuer is to eliminate information associated with this sequence from the router database.

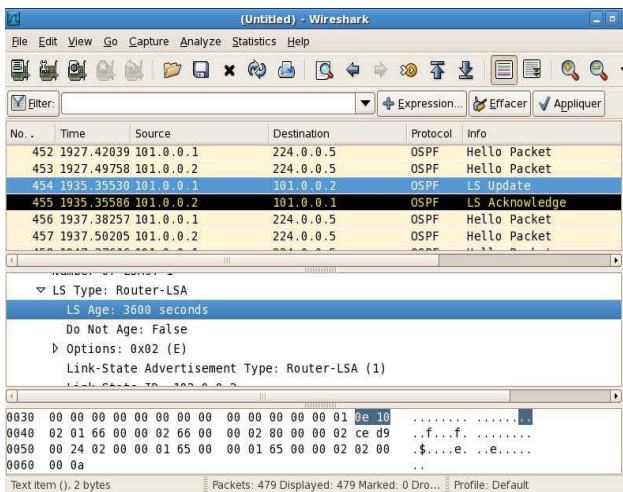


Figure 4. Malicious OSPF packet captured by Wireshark

VII. CONCLUSION

To ensure the good functioning of networks and Internet services, securing routing protocols becomes very essential. It is always necessary to diversify security tools and to propose solutions suitable to new attacks. As a complementary security tool, Honeypots present an interesting means to discover new attacks and study their methods. In the past, this technique was not well explored for the security of routing protocols. For this reason, we propose in this paper a new approach based on

honeypots to protect the routing protocols. Our solution (Honeypot Router) is to deploy a honeypot playing the role of a router. The deployed Honeypot Router helps us to detect new attacks and to learn more about the hacking techniques.

In order to more illustrate our idea, we focus our interest on the RIP, OSPF and BGP protocols. Thus we present the honeypot technology and the routing protocols vulnerabilities. Afterwards, we analyse two attacks against RIP and OSPF protocols. As a future work, we plan to study other in wire and wireless routing protocol. We will also develop a system that automatically gathers and correlates security information collected by different probes installed on the router to simplify the analysis task.

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